

```

board = [" "] * 9

def display():
    print(board[0], "|", board[1], "|", board[2])
    print("-----")
    print(board[3], "|", board[4], "|", board[5])
    print("-----")
    print(board[6], "|", board[7], "|", board[8])

def check():
    win = [(0,1,2), (3,4,5), (6,7,8),
           (0,3,6), (1,4,7), (2,5,8),
           (0,4,8), (2,4,6)]

    for a, b, c in win:
        if board[a] == board[b] == board[c] != " ":
            return True
    return False

player = "X"

for i in range(9):
    display()
    pos = int(input("Player " + player + " position (1-9): ")) - 1

    if board[pos] == " ":
        board[pos] = player

        if check():
            display()
            print(player, "wins!")
            break

        player = "O" if player == "X" else "X"
else:
    display()
    print("Draw!")

```

```

PS C:\Users\BMSCE\Documents\1BM24CS267\week1> python -u "c:\Users\BMSCE\Documents\1BM24CS323\1BM24CS323_WEEK_1_TICTACTOE.py"
| |
-----
| |
-----
| |
Player X position (1-9): 1
X | |
-----
| |
-----
| |
Player O position (1-9): 3
X | | O
-----
| |
-----
| |
Player X position (1-9): 4
X | | O
-----
X | |
-----
| |
Player O position (1-9): 5
X | | O
-----
X | O |
-----
| |

```

```

| |
Player X position (1-9): 7
X | | O
-----
X | O |
-----
X | |
X wins!

```

```

# Vacuum Cleaner - Simple Reflex Agent

def simple_reflex_agent(location, status):
    if status == "Dirty":
        return "Suck"

    if status == "Clean":
        if location == "A":
            return "Move Right"
        else:
            return "Move Left"

location = "A"
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

print("SIMPLE REFLEX VACUUM CLEANER")
print("-----")

for step in range(6):
    status = rooms[location]

    print(f"\nLocation : {location}")
    print(f"Status   : {status}")

    action = simple_reflex_agent(location, status)
    print(f"Action    : {action}")

    if action == "Suck":
        rooms[location] = "Clean"

    elif action == "Move Right":
        location = "B"

    elif action == "Move Left":
        location = "A"

    if rooms["A"] == "Clean" and rooms["B"] == "Clean":
        print("\nBoth rooms are clean.")
        break

```

```

Location : A
Status   : Dirty
Action    : Suck

Location : A
Status   : Clean
Action    : Move Right

Location : B
Status   : Dirty
Action    : Suck

```

```

# Vacuum Cleaner - Goal Based Agent

def goal_based_agent(location, rooms):

    if rooms[location] == "Dirty":
        return "Suck"

    if rooms["A"] == "Dirty":
        if location == "A":
            return "Suck"
        else:
            return "Move Left"

    if rooms["B"] == "Dirty":
        if location == "B":
            return "Suck"
        else:
            return "Move Right"

    return "Stop"

location = "A"

rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

print("GOAL-BASED VACUUM CLEANER")
print("-----")

while True:
    print(f"\nLocation : {location}")
    print(f"Room A   : {rooms['A']}")
    print(f"Room B   : {rooms['B']}")

    action = goal_based_agent(location, rooms)
    print(f"Action    : {action}")

    if action == "Suck":
        rooms[location] = "Clean"

    elif action == "Move Right":
        location = "B"

    elif action == "Move Left":
        location = "A"

    elif action == "Stop":
        print("\nGoal achieved: Both rooms are clean!")
        break

```

GOAL-BASED VACUUM CLEANER

```

Location : A
Room A   : Dirty
Room B   : Dirty
Action    : Suck

```

```

Location : A
Room A   : Clean
Room B   : Dirty
Action    : Move Right

```

```

Location : B
Room A   : Clean
Room B   : Dirty
Action    : Suck

```

```

Location : B
Room A   : Clean
Room B   : Clean
Action    : Stop

```

Goal achieved: Both rooms are clean!